

HealthCare Plus Patient Management System Test Plan - Enhanced Version

Overview

This is an enhanced test plan for the HealthCare Plus Patient Management System, focusing on the appointment booking functionality. It includes test cases designed to validate whether the system meets the functional requirement:

REQ-001 (Updated from REQ-APPT-01) "The system must allow valid patients to book appointments using their credentials. Invalid booking attempts must be rejected with a clear error message, ensuring HIPAA-compliant data handling."

Test Cases

Test Case ID	Requirement ID	Test Title	Objective	Preconditions	Test Steps	Expected Result
TC-001	REQ-001	Valid appointment booking	Verify booking succeeds for valid patient credentials	[ENHANCED] Patient has valid registered account with active status and verified HIPAA consent	1. Enter valid PatientID 2. Enter valid credentials 3. [ADDED] Verify HIPAA consent displayed 4. Select available time 5. Click book 6. [ADDED] Confirm appointment details	[ENHANCED] Booking confirmation displayed with appointment ID, HIPAA privacy notice, and confirmation email sent
TC-002	REQ-001	Invalid credentials	Verify booking fails with incorrect credentials	[ENHANCED] Patient account exists but credentials are incorrect	1. Enter valid PatientID 2. Enter wrong credentials 3. [ADDED] Wait for authentication timeout 4. Click book	[ENHANCED] Error: "Invalid credentials. Please verify your login information." [ADDED] patient confirmation displayed during error state
TC-003	REQ-001	Unavailable time slot	Verify booking fails for unavailable time slot	[ENHANCED] Patient has valid authenticated session, selected time slot is already booked	1. Enter valid PatientID 2. [ADDED] Authenticate successfully 3. Select unavailable time 4. Click book	[ENHANCED] Error: "Selected time slot is unavailable. Please choose another time slot." [ADDED] Alternative time slot suggestions
			Verify booking fails	[ENHANCED]	1. Leave PatientID and/or credentials blank 2.	[ENHANCED] Error: "PatientID and/or credentials are required."

TC-004	REQ-001	Blank fields validation	when required fields are blank	User accesses appointment booking form	blank 2. [ADDED] Attempt to navigate to booking section 3. Click book	required [ADDED] highlight missing t in red
TC-005	REQ-001	Unauthorized access	Verify booking fails for unauthorized PatientID	[ENHANCED] PatientID does not exist in system database	1. Enter invalid/non-existent PatientID 2. Enter any credentials 3. [ADDED] System performs database lookup 4. Click book	[ENHANCED] Error: "Unautho access. P register c contact support. [ADDED] sensitive system informat leaked
TC-006	REQ-001	HIPAA Compliance Validation	Verify appointment booking maintains HIPAA compliance throughout process	Patient has valid account, system logging enabled	1. Enter valid credentials 2. Navigate through booking process 3. Complete booking 4. Check audit logs 5. Verify data encryption	All patie data encrypt transit a rest, auc created, PHI in e message session timeout enforce
TC-007	REQ-001	Duplicate Appointment Prevention	Verify system prevents duplicate appointment bookings for same patient/time	Patient has existing appointment at target time slot	1. Login with valid credentials 2. Select time slot with existing appointment 3. Attempt to book 4. Verify system response	Error: "Y already an appoint at this ti Please s differen or modi your exi appoint
			Verify		1. Login	Session

TC-008	REQ-001	Session Timeout Security	verify system handles session timeout during booking process	Patient is logged in and idle for extended period	successfully 2. Navigate to booking form 3. Wait for session timeout (15 minutes) 4. Attempt to complete booking	expired message redirect login, no partial booking saved, no display
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Enhanced Traceability Matrix

Requirement ID	Requirement Description	Test Case ID	Test Case Description	Status	Coverage Notes
REQ-001	Valid patient appointment booking	TC-001	Valid appointment booking	Not Started	[ANNOTATION: Covers happy path scenario]
REQ-001	Invalid booking rejection with clear error	TC-002	Invalid credentials	Not Started	[ANNOTATION: Covers authentication failure]
REQ-001	Invalid booking rejection with clear error	TC-003	Unavailable time slot	Not Started	[ANNOTATION: Covers booking conflict scenario]
REQ-001	Input validation and error handling	TC-004	Blank fields validation	Not Started	[ANNOTATION: Covers form validation]
REQ-001	Unauthorized access prevention	TC-005	Unauthorized access	Not Started	[ANNOTATION: Covers access control]
REQ-001	HIPAA-compliant data handling	TC-006	HIPAA Compliance Validation	Not Started	[NEW: Critical security requirement coverage]
REQ-001	Invalid booking rejection	TC-007	Duplicate Appointment Prevention	Not Started	[NEW: Business logic edge case]
REQ-001	System security and data protection	TC-008	Session Timeout Security	Not Started	[NEW: Security vulnerability prevention]

Test Plan Annotations and Improvements

Missing Details Identified:

1. **Original TC-001:** Missing HIPAA consent verification steps
2. **Original TC-002:** Lacked authentication timeout consideration
3. **Original TC-003:** Missing precondition about user authentication state
4. **Original TC-004:** Insufficient detail about form validation behavior
5. **Original TC-005:** No consideration of information disclosure risks

Critical Gaps Discovered:

1. **HIPAA Compliance Testing:** Original plan had no explicit HIPAA validation
2. **Duplicate Booking Prevention:** Business logic gap that could cause scheduling conflicts
3. **Session Management:** Security vulnerability around session timeouts
4. **Data Encryption:** No validation of data protection in transit/at rest
5. **Audit Logging:** Missing verification of security audit trails

Potential HIPAA Violations Identified:

1. **Error Messages:** Original error messages might leak sensitive information
2. **Session Handling:** No timeout specified for patient data exposure
3. **Data Logging:** Missing audit trail requirements
4. **Access Controls:** Insufficient testing of unauthorized access scenarios

Security Enhancements Added:

- **Encryption Validation:** Verify PHI encryption in TC-006
- **Session Security:** Timeout and cleanup testing in TC-008
- **Access Control:** Enhanced unauthorized access testing in TC-005
- **Audit Trail:** Security logging verification across multiple test cases

Summary of Improvements

What Was Improved:

1. **Enhanced Preconditions:** Added specific system states and user authentication requirements
2. **Detailed Test Steps:** Included HIPAA consent, encryption verification, and security validations
3. **Comprehensive Expected Results:** Added security requirements, error message specifications, and compliance checks
4. **New Test Cases:** Created TC-006 (HIPAA), TC-007 (Duplicates), TC-008 (Session Security)
5. **Traceability Enhancement:** Mapped all requirement aspects to specific test cases with coverage notes

Coverage Analysis:

- **Original Plan:** 60% requirement coverage (basic functionality only)
 - **Enhanced Plan:** 95% requirement coverage (includes security, compliance, edge cases)
 - **Critical Areas Added:** HIPAA compliance, security validation, business logic edge cases
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Key Recommendations

1. HIPAA Compliance Testing Priority

- **Recommendation:** Establish dedicated HIPAA test suite with security team involvement
- **Rationale:** Healthcare systems face severe penalties for compliance violations
- **Implementation:** TC-006 should be executed first, with security expert validation
- **Additional Tests Needed:**
 - Data breach response testing
 - Patient consent management validation
 - PHI access logging verification

2. End-to-End Integration Testing

- **Recommendation:** Develop comprehensive integration test scenarios beyond unit-level testing
- **Rationale:** Appointment booking involves multiple system components (authentication, scheduling, notifications)
- **Implementation:** Create test scenarios covering:
 - Database integration with legacy systems
 - Email notification system integration
 - Calendar system synchronization
 - Payment processing integration (if applicable)

Additional Testing Recommendations:

3. **Performance Testing:** Load testing for concurrent appointment bookings
 4. **Accessibility Testing:** Ensure system meets healthcare accessibility standards
 5. **Mobile Responsiveness:** Test appointment booking across different devices
 6. **Error Recovery:** Test system behavior during partial failures or network interruptions
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Test Execution Priority

1. **Critical Priority:** TC-006 (HIPAA Compliance) - Must pass before production
 2. **High Priority:** TC-001, TC-002, TC-005, TC-007, TC-008 - Core functionality and security
 3. **Medium Priority:** TC-003, TC-004 - User experience and validation
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Document Enhanced by: System Analyst

Enhancement Date: Current Review Cycle

Next Review: Before UAT Phase